ASSIGNMENT 4 / INTRODUCTION TO JAVASCRIPT

Interactive Web Features

Team: Web Dev Guide
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Group: SE-2514
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Objective

Build and explain five interactive features with JavaScript: input validation, an FAQ accordion, a popup form, background color changes, and a live date and time display.

Project result

A dedicated **JavaScript Interactive Lab** page expanded the Web Dev Guide to seven pages. The new page is linked from every previous page. HTML, CSS, and JavaScript are in separate files; all page footers list the team members.

01 Form validation	**02 FAQ accordion**
03 Popup form	**04 Background color**
05 Live date & time	**Responsive layout**

Website URL

Published project: https://e1noor.github.io/front_end/
Interactive Lab: https://e1noor.github.io/front_end/7-js-interactive.html

Forms are demonstration features. No data is transmitted or saved.

Tasks 1 and 2

Task 1 - Form validation

A registration form checks required values and gives a specific message next to each incorrect field.

1. Added name, email, password, and confirmation fields with labels.

2. JavaScript checks empty fields, email pattern, a minimum of eight password characters, and matching passwords.

3. The submit event is prevented; errors appear next to fields, focus moves to the first error, and valid input receives success feedback.

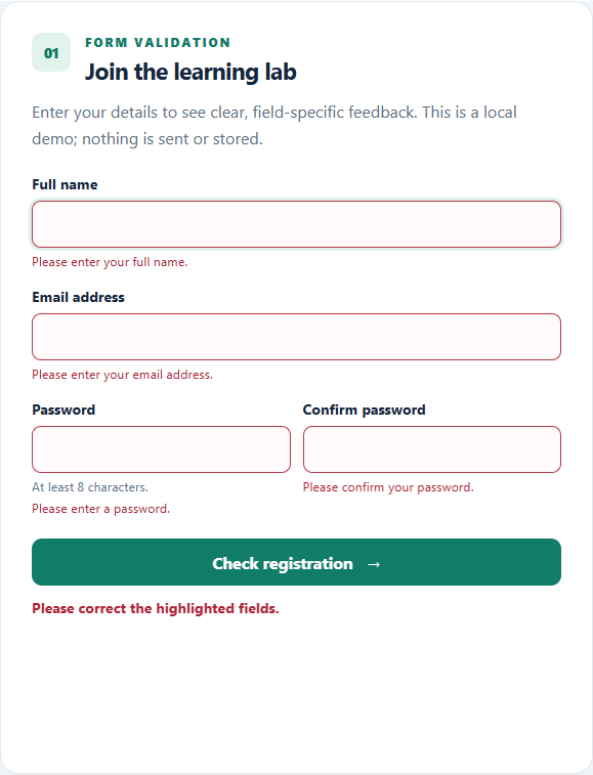


Figure 1. Missing fields show clear errors.

Task 2 - FAQ accordion

Four questions are hidden by default and can be opened or closed individually.

1. Created question buttons and answer regions in HTML.

2. On click, JavaScript changes aria-expanded and the answer height.

3. A CSS height transition animates both opening and closing.

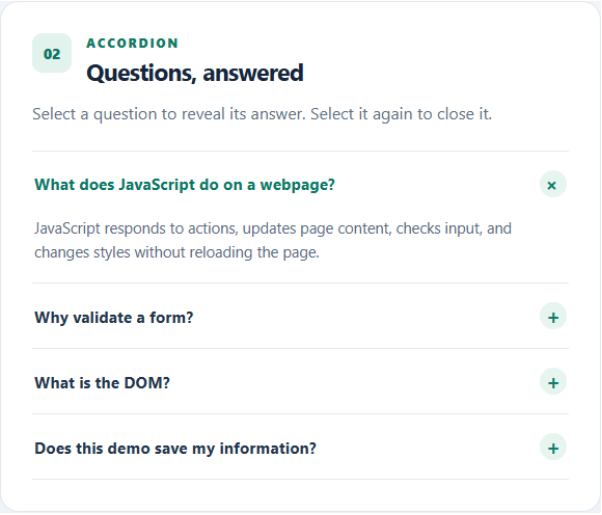


Figure 2. An expanded answer.

Tasks 3, 4, and 5

Task 3 - Popup subscription form

The subscription form starts hidden and opens from either of two buttons.

1. Show the dialog and move focus into its email field.
2. Validate the email locally and show feedback.
3. Close via the X button, Escape key, or the backdrop; return focus to the opening button.

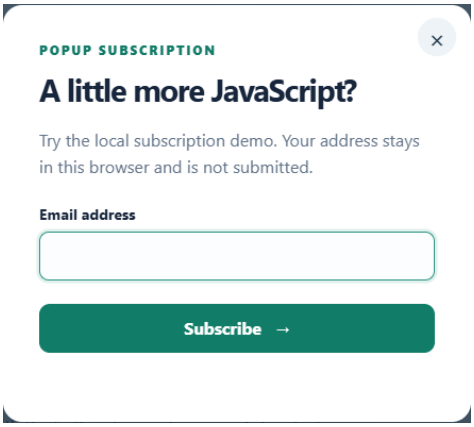


Figure 3. Open subscription popup.

Task 4 - Change background color

A button cycles through Cloud, Mint, Peach, and Lavender.

1. Store color names and values in a JavaScript array.
2. Advance the index on click and set the body background color.
3. Update the visible color label; CSS supplies a smooth transition.

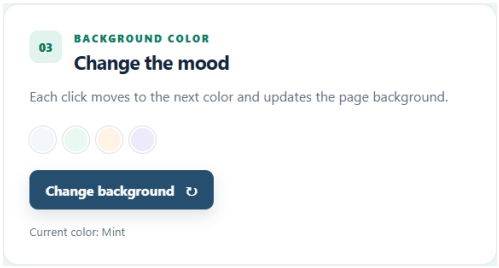


Figure 4. Mint background selected.

Task 5 - Current date and time

The time block shows the device local date and time in a readable format.

1. Create a Date object and format it with Intl.DateTimeFormat.
2. Update the visible time immediately and every second.
3. Set the time element dateTime value to the ISO timestamp.

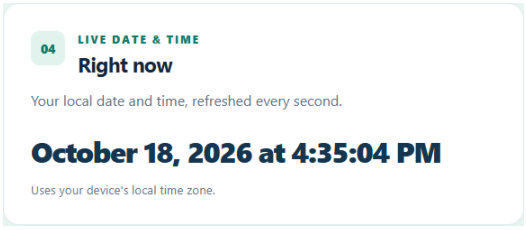


Figure 5. Live date and time display.

Verification and reflection

Browser checks

The page was opened in Microsoft Edge and Google Chrome at desktop (1440 px) and mobile (390 px) widths. Automated browser checks covered empty and malformed form data, successful registration feedback, accordion opening and closing, all popup closing methods, subscription email feedback, background color change, and the live clock. No JavaScript page errors or mobile horizontal overflow were detected.

Files and deployment

Main page: 7-js-interactive.html
Styles: css/interactive.css
Script: js/interactive.js
Published URL: https://e1noor.github.io/front_end/7-js-interactive.html

Reflection

This assignment shows how event listeners connect user actions to DOM updates. The most important lesson was to pair each visual state change with clear feedback: errors identify the exact field, accordion buttons expose their state, the popup returns focus after closing, and the selected background color is named. The features work without a backend, so the forms are appropriate as interactive demonstrations rather than real registration or subscription services.

Defense preparation

Each team member should be able to explain event.preventDefault(), DOM element selection, aria-expanded, modal close events, array-based color cycling, Date/Intl formatting, and the difference between client-side validation and server-side processing.

Mobile layout evidence

The responsive page stacks cards into one column, wraps the navigation, and keeps every control reachable on a narrow viewport.

